Project: Worked examples
Title: Load combinations (EN 1990)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the design values of one action effect under every EN 1990 Annex A1 combination for buildings, from persistent to accidental. Each variable action leads in turn; the envelopes and governing combinations are reported.

WHAT IT CALCULATES	RELATION
ULS 6.10	$\Sigma\ \gamma_G{\cdot}G + \gamma_Q{\cdot}K_{FI}{\cdot}Q_1 + \Sigma\ \gamma_Q{\cdot}K_{FI}{\cdot}\psi_0{\cdot}Q_i$ (K_{FI} also on $\gamma_{G,sup}$)
ULS 6.10a	$\Sigma\ \gamma_G{\cdot}G + \Sigma\ \gamma_Q{\cdot}K_{FI}{\cdot}\psi_0{\cdot}Q_i$; 6.10b: $\Sigma\ \xi{\cdot}\gamma_G{\cdot}G + \gamma_Q{\cdot}K_{FI}{\cdot}Q_1 + \Sigma\ \gamma_Q{\cdot}K_{FI}{\cdot}\psi_0{\cdot}Q_i$
Permanent factor	$\gamma_{G,sup}$ when the action adds to the effect sought, $\gamma_{G,inf}$ when it opposes it
Variable actions	included only when they add to the effect sought (otherwise zero)
SLS characteristic 6.14b	$\Sigma\ G + Q_1 + \Sigma\ \psi_0{\cdot}Q_i$; frequent 6.15b: $\Sigma\ G + \psi_1{\cdot}Q_1 + \Sigma\ \psi_2{\cdot}Q_i$; quasi-permanent 6.16b: $\Sigma\ G + \Sigma\ \psi_2{\cdot}Q_i$
Accidental 6.11b	$\Sigma\ G + A_d + \psi_1{\cdot}Q_1 + \Sigma\ \psi_2{\cdot}Q_i$

Basis: Combinations for persistent or transient design situations (6.10, 6.10a, 6.10b)[1](cl. 6.4.3.2); Combinations for accidental design situations (6.11b)[1](cl. 6.4.3.3); Serviceability combinations (6.14b, 6.15b, 6.16b)[1](cl. 6.5.3); Application for buildings: ψ and γ factors[1](Annex A1, Tables A1.1, A1.2(B)); UK National Annex: UK values of partial factors[1](Table NA.A1.2(A) to (C)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
actions	Actions (one effect, signed)	2 row(s)		drawing: GA-L2-001 rev C	any
unit	Unit of the effect	kN/m		assumption: Line load on the beam.	any
method	ULS expression	6.10a and 6.10b		spec: UK NA Table NA.A1.2(B): the less favourable of 6.10a and 6.10b	any
$\gamma_{G,sup}$	$\gamma_{G,sup}$ (unfavourable permanent)	1.35		spec: UK NA Table NA.A1.2(B)	1 … 2
$\gamma_{G,inf}$	$\gamma_{G,inf}$ (favourable permanent)	1		spec: UK NA Table NA.A1.2(B)	0.8 … 1.2
γ_Q	γ_Q (variable)	1.5		spec: UK NA Table NA.A1.2(B)	1 … 2
ξ	ξ reduction for 6.10b	0.925		spec: UK NA Table NA.A1.2(B)	0.8 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
K_{FI}	Consequence-class factor	1		assumption: Consequence class CC2.	0.9 … 1.1

TABLE ROW	VALUES
actions[1]	name = Slab, finishes and services; category = G permanent; F_k = 18
actions[2]	name = Office imposed; category = B office; F_k = 10

3. Assumptions

A1. Each action's characteristic value is its signed contribution to one scalar effect. (Linear superposition (EN 1990 cl. 6.3.1 representative values).)
A2. The combinations are those of EN 1990 Annex A1 for buildings. (EN 1990 Annex A1 scope.)
A3. Bridge, fatigue and seismic combinations are not included. (EN 1990 Annex A1 scope.)
A4. Without a ψ record the recommended values of EN 1990 Annex A1 are used and labelled on the sheet. (No National Annex table is shipped.)
A5. UK National Annex values of ψ and γ come from a user record. (No National Annex table is shipped.)
A6. Each permanent action takes $\gamma_{G,sup}$ when it adds to the effect sought and $\gamma_{G,inf}$ when it opposes it. (EN 1990 Table A1.2(B) note on favourable and unfavourable permanent actions.)
A7. For an equilibrium check with one source, the source's permanent actions are entered as one row. (EN 1990 Table A1.2(B) note on favourable and unfavourable permanent actions.)
A8. unit (Unit of the effect) is taken as kN/m. (Line load on the beam.)
A9. K_{FI} (Consequence-class factor) is taken as 1. (Consequence class CC2.)
A10. ψ factors: EN 1990 Annex A1 recommended values (Table A1.1). (No ψ record entered.)

4. Working

[8] $C1 = \gamma_{G,sup} \cdot K_{FI} \cdot G1 + \gamma_Q \cdot K_{FI} \cdot \psi_0 \cdot Q2$ [1](cl. 6.4.3.2)
$= 1.35 \times 18 + 1.05 \times 10$
$= 34.8\ kN/m$
ULS 6.10a (max)

[9] $C2 = \xi \cdot \gamma_{G,sup} \cdot K_{FI} \cdot G1 + \gamma_Q \cdot K_{FI} \cdot Q2$ [1](cl. 6.4.3.2)
$= 1.2488 \times 18 + 1.5 \times 10$
$= 37.478\ kN/m$
ULS 6.10b, Office imposed leading (max)

[10] $C3 = 1 \cdot G1 + 1 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 1 \times 10$
$= 28\ kN/m$
SLS characteristic 6.14b, Office imposed leading (max)

[11] $C4 = 1 \cdot G1 + \psi_1 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 0.5 \times 10$
$= 23\ kN/m$
SLS frequent 6.15b, Office imposed leading (max)

[12] $C5 = 1 \cdot G1 + \psi_2 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 0.3 \times 10$
$= 21\ kN/m$
SLS quasi-permanent 6.16b (max)

[13] $D1 = \gamma_{G,inf} \cdot G1$ [1](cl. 6.4.3.2)
$= 1 \times 18$
$= 18\ kN/m$
ULS 6.10a (min)

[14] D2 = $\gamma_{G,inf} \cdot G1$ [1](cl. 6.4.3.2)
= 1×18
= 18 kN/m
ULS 6.10b, no variable action (min)
[15] D3 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS characteristic 6.14b, no variable action (min)
[16] D4 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS frequent 6.15b, no variable action (min)
[17] D5 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS quasi-permanent 6.16b (min)
[18] $E_{d,max}$ = maximum over the combinations
= ULS 6.10b, Office imposed leading
= 37.478 kN/m
ULS design effect, maximum
[19] $E_{d,min}$ = minimum over the combinations
= ULS 6.10a
= 18 kN/m
ULS design effect, minimum
[20] $E_{k,max}$ = maximum over the combinations
= SLS characteristic 6.14b, Office imposed leading
= 28 kN/m
SLS characteristic effect, maximum
[21] $E_{k,min}$ = minimum over the combinations
= SLS characteristic 6.14b, no variable action
= 18 kN/m
SLS characteristic effect, minimum
[22] $E_{fr,max}$ = maximum over the combinations
= SLS frequent 6.15b, Office imposed leading
= 23 kN/m
SLS frequent effect, maximum
[23] $E_{fr,min}$ = minimum over the combinations
= SLS frequent 6.15b, no variable action
= 18 kN/m
SLS frequent effect, minimum
[24] $E_{qp,max}$ = maximum over the combinations
= SLS quasi-permanent 6.16b
= 21 kN/m
SLS quasi-permanent effect, maximum
[25] $E_{qp,min}$ = minimum over the combinations
= SLS quasi-permanent 6.16b
= 18 kN/m
SLS quasi-permanent effect, minimum

5. Results

SYMBOL	VALUE	DESCRIPTION
$E_{d,max}$	37.48 kN/m	ULS design effect, maximum
$E_{d,min}$	18 kN/m	ULS design effect, minimum

SYMBOL	VALUE	DESCRIPTION
$E_{k,max}$	28 kN/m	SLS characteristic effect, maximum
$E_{qp,max}$	21 kN/m	SLS quasi-permanent effect, maximum
$E_{d,gov}$	"ULS 6.10b, Office imposed leading"	ULS design effect, governing (max)
$E_{k,min}$	18 kN/m	SLS characteristic effect, minimum
$E_{k,gov}$	"SLS characteristic 6.14b, Office imposed leading"	SLS characteristic effect, governing (max)
$E_{fr,max}$	23 kN/m	SLS frequent effect, maximum
$E_{fr,min}$	18 kN/m	SLS frequent effect, minimum
$E_{fr,gov}$	"SLS frequent 6.15b, Office imposed leading"	SLS frequent effect, governing (max)
$E_{qp,min}$	18 kN/m	SLS quasi-permanent effect, minimum
$E_{qp,gov}$	"SLS quasi-permanent 6.16b"	SLS quasi-permanent effect, governing (max)

6. Checks (governing first)

None.

7. Applicability

WARNING records: tabulated values not taken from a reference record are $\gamma_{G,sup}$, $\gamma_{G,inf}$, γ_Q, ξ (EN 1990 partial and ψ factors). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

`Template struct.load-combinations@1.2.0 Content hash 2cdaa6adaf7b0cf8 Engine calc-engine/1.0.0`
`Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. British Standards Institution. Eurocode: Basis of Structural Design. BS EN 1990. BSI.